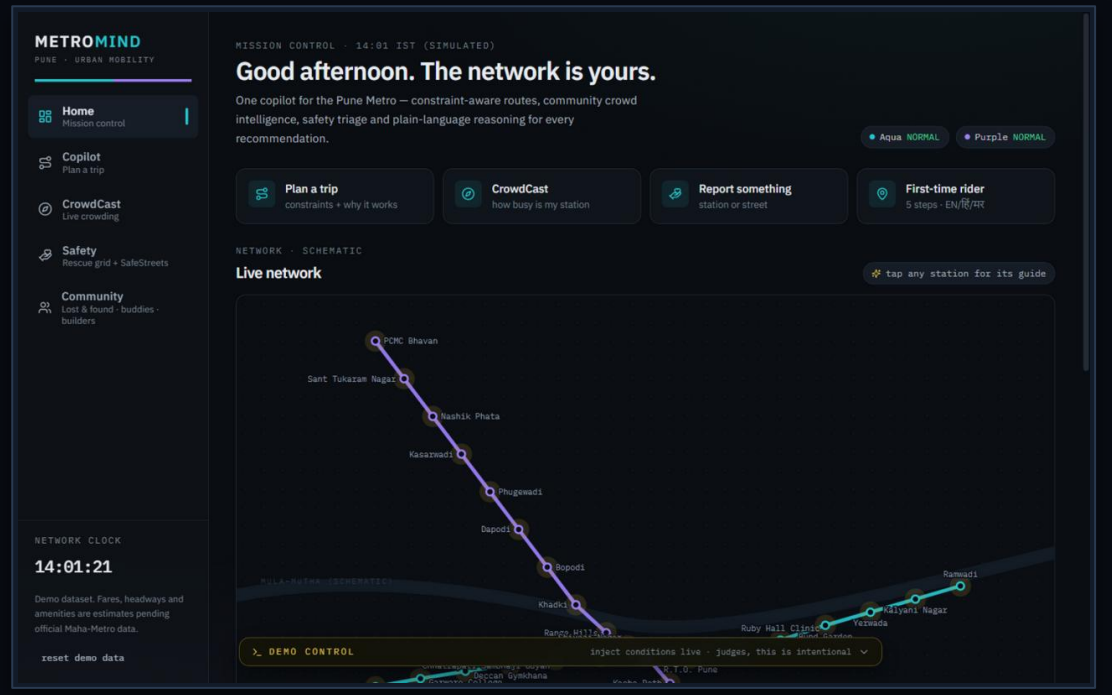
CODEX DAYLINE · URBAN MOBILITY TRACK · PUNE METRO BUILD SPRINT

METROMIND

Pune Moves. MetroMind Thinks Ahead.

An AI commute copilot that plans constraint-aware journeys, reasons over crowd reports and injected disruptions, explains every recommendation in plain language — and pairs it with safety triage, accessibility guidance and commuter community tools.

problem → solution → live demo → impact



ACTUAL BUILD · MISSION CONTROL — LIVE NETWORK, CROWD HALOS, DEMO CONTROL BAR

DEMO DATA THROUGHOUT — NO REAL-TIME FEED CLAIMED

TEAM URBANSYNC

Arya Jannawar · Siddhant Pardeshi · Avani Jannawar

github.com/Arya-jannawar/MetroMind

A Journey Is More Than a Route.

A route suggestion answers “where”. A commuter also has to answer when, how much, how crowded — and what to do when one of those changes mid-journey.

01

First-time riders

Which line, which exit, which gate? Station knowledge is scattered across signs, apps and word-of-mouth.

02

Daily commuters

How crowded will the next platform be? There is no way to see it before you commit to a departure.

03

When something changes

A delay, a spike, a missed train — no single place explains what changed and what to do now.

WHAT THE RIDER IS REALLY BALANCING

“Reach Deccan by 09:30.” — a deadline, not a preference

“Stay under ₹50.” — a budget the plan must respect

“Avoid packed trains.” — comfort the route app ignores

“Step-free, if possible.” — accessibility needs

“Aqua line delayed 15’?” — the plan breaks — then what?

Qualitative, honest framing — no ridership statistics are claimed.



One Copilot. Multiple Travel Decisions.

STEP 1

USER INPUT

"Reach Deccan by 9:30. Budget ₹50. Avoid crowds."

stated as a goal, not two station names

STEP 2

DECISION ENGINE

Constraint-aware Dijkstra over the demo network graph — ride time, fare, transfers, crowd penalty.

runs on-device, no API key needed

STEP 3

EXPLAINED PLANS

Three timing strategies, each with fare, crowd forecast and a one-sentence why.

trade-offs visible, not hidden

STEP 4

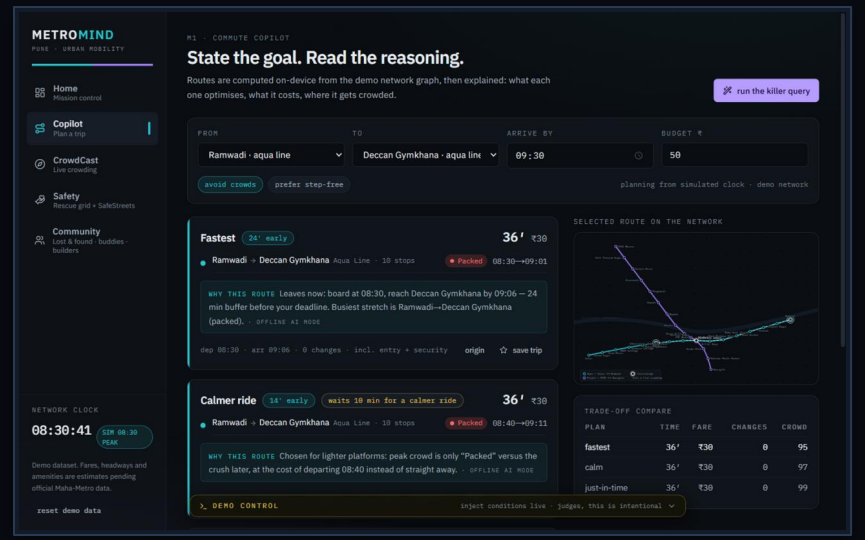
ADAPTIVE REPLY

Inject a delay or a crowd spike — every card recomputes and explains the change.

the demo moment, live

- **Constraint chips, not forms** arrive-by · budget · avoid crowds · step-free — one tap each.
- **Explained, every time** each card carries a "why this route" line in plain language.
- **One shared state core** a crowd report or a delay anywhere re-reasons the whole app.

ACTUAL BUILD



ON-DEVICE ROUTING · EXPLAINABLE BY DEFAULT · AI OPTIONAL

Not Just the Best Route. The Reason Behind It.

KILLER QUERY · RAMWADI → DECCAN GYMKHANA · ARRIVE 09:30 · BUDGET ₹50 — DEMO ESTIMATES

FASTEST

dep 08:30 → arr 09:06

36′ · ₹30 · 0 changes · 10 stops

"Leaves now — 24 min of buffer before the deadline. Busiest stretch: Ramwadi → Deccan Gymkhana (packed)."

CALMER RIDE

dep 08:40 → arr 09:11

36′ · ₹30 · waits 10 min, calmer ride

"Chosen for lighter platforms: peak crowd is 'Packed' versus the crush later — at the cost of departing later."

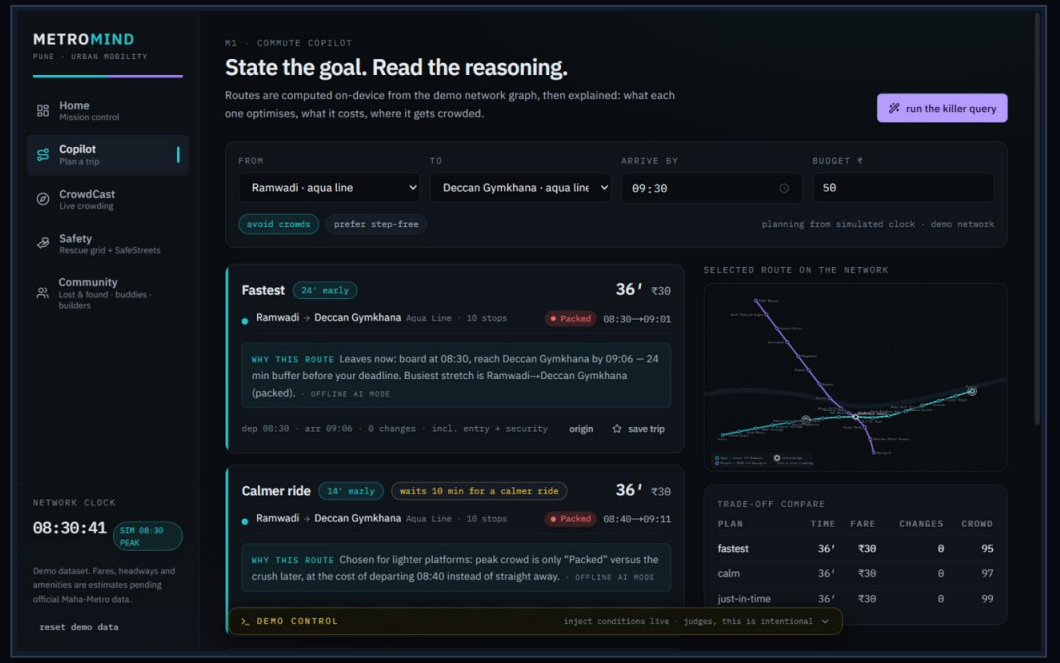
JUST-IN-TIME

dep 08:54 → arr 09:30

36′ · ₹30 · zero buffer

"Arrives exactly at the deadline — the latest safe departure the schedule allows."

Strategies differ in WHEN you travel — honest for a two-line network where the physical path is unique. Off-peak, one honest card is shown instead of padding.



THE DIFFERENTIATOR

MetroMind explains the trade-off instead of returning a route without context — every recommendation ships with its reason.

Five Capabilities. One Connected Platform.

M1 COMMUTE COPILOT

The flagship. Constraint-aware route planning, comparison, plain-language explanations and live rerouting.

- arrive-by · budget · avoid crowds · step-free
- three timing strategies with why-lines
- trade-off compare table
- live disruption re-planning (Demo Control)



M2 CROWDCAST

Station × 15-min crowd heatmap. One-tap anonymous reports — Empty / Moderate / Packed — with 15-minute half-life decay, confidence and freshness.

M3 RESCUE GRID + SAFESTREETS

Incidents ranked by triage score — severity, people affected, footfall, recency — with recommended actions. Street-hazard risk map for the walk in.

M4 STATION GUIDE / ACCESSWAY

5-step first-rider guide in EN / हिंदी / मराठी. Per-station pages: exits, parking, accessibility, first/last mile, departures, crowding now.

M5 COMMUNITY CONNECT

Lost & found board with search, travel-buddy matching with a why-matched reason, ConnectIQ-style builder matching, station events.

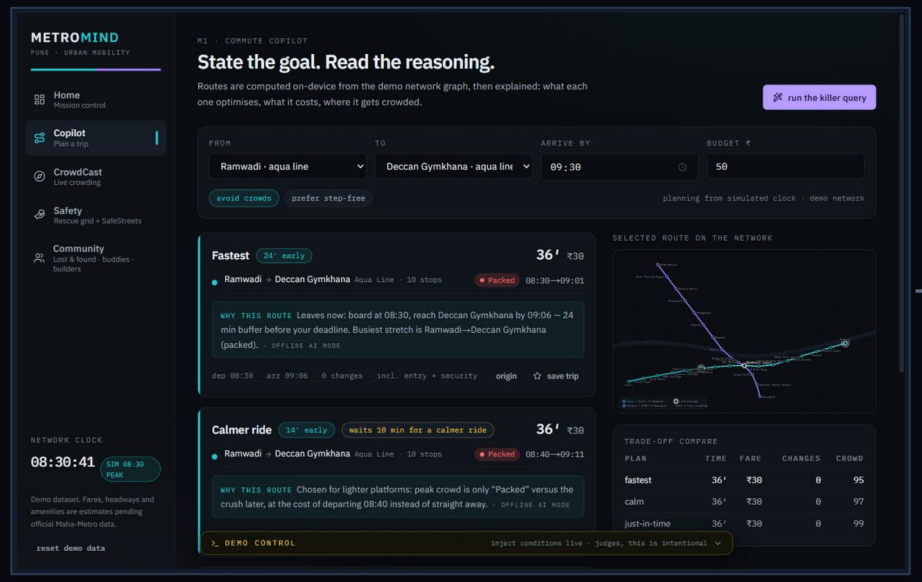
SHARED METRO-DATA + DISRUPTION-STATE CORE (lib/), persisted locally. A report anywhere re-reasons everywhere.

29 stations · Aqua + Purple lines · District Court interchange — crowd reports, disruptions and incidents in one engine



When the Metro Changes, the Recommendation Changes.

BEFORE — three explained plans, Fastest recommended



DEMO CONTROL BAR

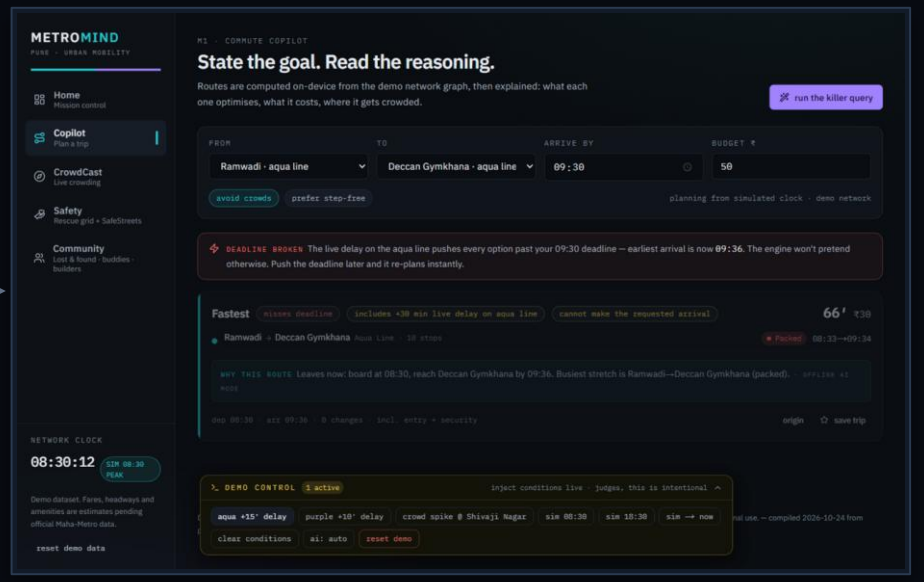
▸ **aqua +15′ delay**

repeat press → +30′ total

↓

cards recompute live, in front of the judges

AFTER — "Deadline broken": earliest arrival 09:36



Recomputed live

+30′ chips, updated arrivals, re-ranked plans — no cutaway, no refresh.

Honest failure state

"Earliest arrival is now 09:36 — the engine won't pretend otherwise."

Relax the deadline

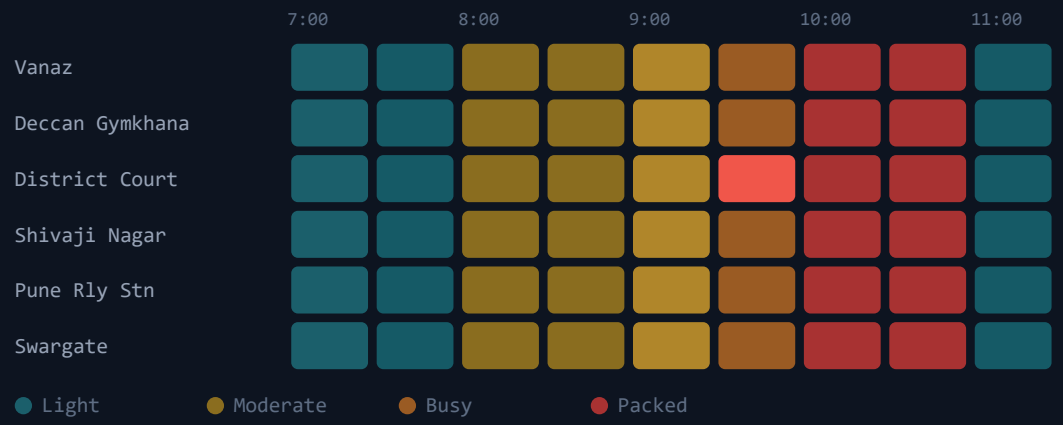
push 09:30 later → instant re-plan. The demo is interactive, not a recording.

SIMULATED DEMO CONDITIONS — injected by the built-in Demo Control bar. No real-time Maha-Metro feed is claimed.

Small Reports. Smarter Decisions.

M2 · CROWDCAST

Station × time crowd heatmap



- One-tap anonymous reports: Empty / Moderate / Packed — no account.
- 15-min half-life decay · confidence + freshness shown on every report.
- One tap re-badges Copilot route advice across the whole app.

M3 · RESCUE GRID + SAFESTREETS

Help gets ranked, not just logged

#1 LOST CHILD · Swargate score 100/100

why: type +88 · 2 affected +10 · hub footfall +10 · just now +8

RECOMMENDED Freeze exits 10 min · announce on PA · review CCTV · staff lift lobbies

- Triage score = severity + people affected + station footfall + recency — the queue re-ranks and explains itself.
- Assign team / resolve workflow in the prototype; SafeStreets adds a street-hazard risk map for the walk to the station.

PROTOTYPE ONLY — does not dispatch real emergency services. In a real emergency in India, call 112 or notify metro staff. Crowd reports and incidents are seeded samples labelled “sample” / “DEMO DATA” in the UI.

Built Around a Shared Decision Engine.

INTERFACE

Next.js 15 App Router · React 19 · TypeScript · Tailwind CSS — 7 routes: home · copilot · crowd · safety · guide · community · station/[id]

↓

STATE

Zustand stores × 6 (live · crowd · rescue · community · safe · prefs) — persisted to localStorage; "reset demo" wipes every key

↓

SHARED DATA CORE

data/pune-metro.json — 29 stations · Aqua + Purple · District Court interchange + seeded reports, incidents, hazards — labelled DEMO DATASET

↓

ENGINE — lib/ · PURE TYPESCRIPT, ZERO FRAMEWORK IMPORTS

Dijkstra over (station, line) states · fare ₹10 + ₹2/stop, cap ₹35 · crowd penalty + 15-min half-life decay · triage scoring · explain layer — covered by 28 vitest tests

↓

OPTIONAL AI LAYER

/api/copilot → GLM via z.ai, server-side key only · 3 s timeout · strict-JSON validation → on any failure: deterministic template explanations, badged "offline AI mode"

RUNS WITH NO API KEY

Routing, crowd, triage and the full demo are complete in deterministic mode. AI is an upgrade, never a dependency.

28 TESTS · NEXT BUILD CLEAN

Routing constraints, fare capping, crowd decay, triage ranking, persistence. Zero console errors on desktop + 390 px.

PWA-READY SHELL

Manifest shipped; offline trip caching is on the roadmap — honestly marked as future work.

Engine band is the moat: pure TypeScript, framework-free, unit-tested — portable to a real feed without rewriting the app.

From Route Search to Decision Support.

CAPABILITY	TYPICAL ROUTE PLANNER	METROMIND — BUILT
Route suggestions	Varies — usually a single best route	Three explained plans with visible trade-offs
Travel constraints	Varies — rarely combined	Arrive-by · budget · avoid crowds · step-free, one tap each
Disruption response	Varies	Injected delays re-plan live — with honest failure states
Crowd intelligence	Varies	Community heatmap: one-tap reports, decay, confidence
Safety support	Varies	Triage-ranked incident queue with recommended actions
Community tools	Varies	Lost & found · travel buddies · builder matching · events

Comparisons are fair by design — "varies" means capable apps exist; the claim is the combination, not that these are unique.

The combination is the differentiator: one shared state core means a crowd report, an injected delay and an incident all change the same recommendation surface — in front of the judge, live.

Designed for the Realities of Urban Commuting.

01
Informed decisions

Time, fare and crowd trade-offs visible before you commit to a departure.

02
Visible crowding

Fresh, decaying community reports give platforms a forecast they never had.

03
Accessible guidance

Step-free chips, lift and exit details, first-rider guide in EN / हिंदी / मराठी.

04
Structured civic voice

Hazards and incidents ranked and explained — not lost in a comment feed.

A PRACTICAL PATH FORWARD — PHASES AS STATIONS ON ONE LINE

PHASE 1 · NOW

Demo on a verified station list with labelled simulated data — this build.

PHASE 2

Validate with commuters; verify station facilities and accessibility on site.

PHASE 3

Explore authorized metro-data integrations — GTFS / Maha-Metro — for real pilots.

PHASE 4

Evaluate usefulness, reliability, accessibility and privacy before wider deployment.

Presented as potential benefits and future steps — not measured outcomes. No adoption or accuracy numbers are claimed.

A Focused Prototype With a Practical Path Forward.

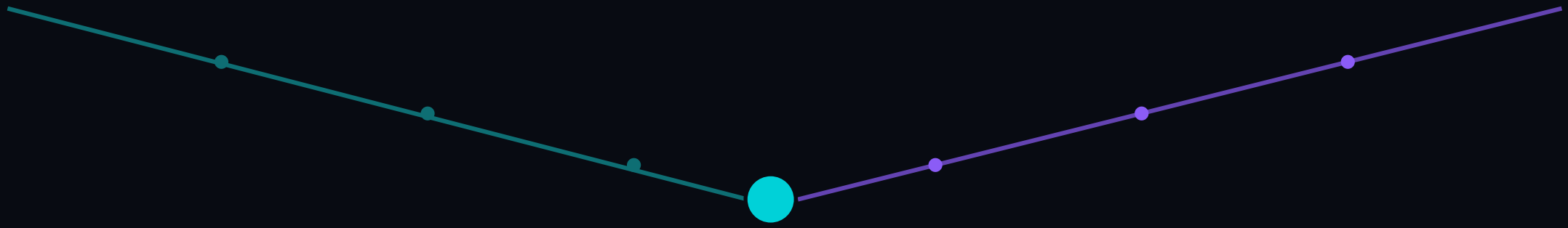
BUILT & VERIFIED IN THE REPO

- ✓ Constraint-aware copilot: Dijkstra engine, 3 strategies, why-lines, trade-off table
- ✓ Demo Control: inject delays (+15′ → +30′ accumulate), crowd spikes, simulated clock
- ✓ CrowdCast: heatmap, one-tap reports, 15-min half-life decay, confidence + freshness
- ✓ Rescue Grid: triage scoring + recommended actions · SafeStreets street-hazard map
- ✓ Station Guide: 5-step first-rider guide EN / हिंदी / मराठी + per-station pages
- ✓ Community: lost & found, buddy match with reasons, ConnectIQ-style builder matching
- ✓ AI upgrade path: GLM via z.ai, 3 s timeout, instant deterministic fallback
- ✓ Quality gate: 28 vitest tests · clean build · zero console errors (desktop + mobile)

NEXT — IN DELIVERY ORDER

- ▸ Verified official data; authorized Maha-Metro / GTFS feed integration
- ▸ AFC-gate-based crowding to replace or calibrate community reports
- ▸ Authority dashboards for the Rescue Grid (needs operator sign-off)
- ▸ Full app-wide Hindi / Marathi i18n beyond the guide
- ▸ PWA offline trip caching; multimodal PMPML / BRTS routing
- ▸ Photo verification for SafeStreets; abuse-prevention for reports
- ▸ Real commuter testing, accessibility validation, production deployment

HONESTY POLICY Every fare, headway and amenity is labelled DEMO DATASET / ESTIMATE in-product. Simulated information is never presented as official Maha-Metro operational data.



METROMIND

Every Commute Deserves a Smarter Decision.

Route planning, explainable AI, crowd intelligence, safety guidance and community support — one commuter-focused experience, built and demo-ready.

BUILT FOR PUNE. DESIGNED FOR SMARTER JOURNEYS.

Thank you. Questions?

TEAM URBANSYNC — ARYA JANNAWAR · SIDDHANT PARDESHI · AVANI JANNAWAR